

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

When the engine is operating with the rocker lever covers removed, oil splashing can cause personal injury. To reduce the possibility of personal injury, wear eye and face protection.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the air crossover. Refer to Procedure 010-020 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-020.html)

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- Disconnect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)

- Remove the injector exhaust gas temperature sensor (EGTS) harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)

Natural Gas

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

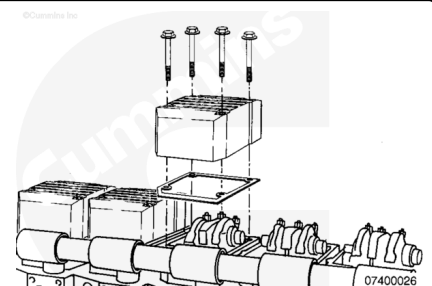
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the spark plug wire (non shielded). Refer to Procedure 013-014 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-014.html)
- Remove the upper portion of the air intake manifold lateral and vertical clamping brackets, if required.

Remove

with Mechanically Actuated Injector

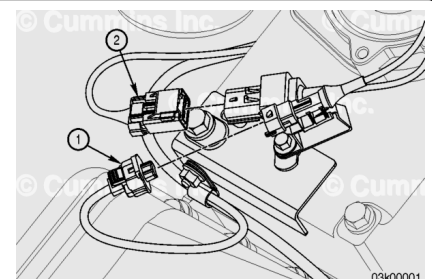
Crankcase breathers can be mounted in any cylinder location. Record the number of breathers and their correct location before removal.

Remove the rocker lever cover and discard the gasket.

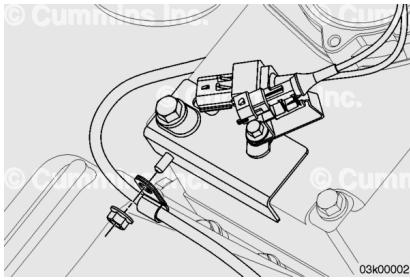


with Electronically Actuated Injector

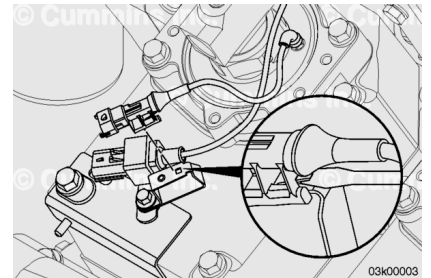
Disconnect the electrical connectors for the injector (1) and exhaust gas temperature sensor (2).



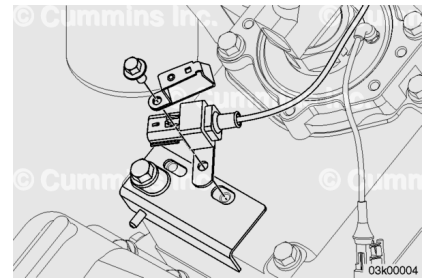
Remove the P-clip securing the wiring harness to the rocker lever cover.



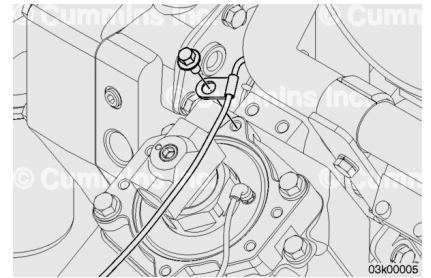
Remove the injector wiring harness from the rocker cover bracket by using a small flat tip screw driver to release the locking mechanism.



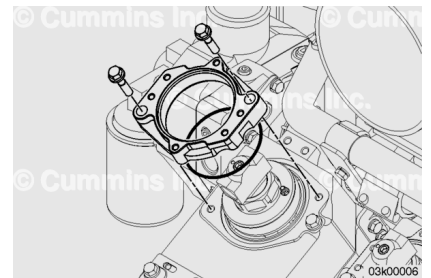
Remove the injector wiring harness bracket and exhaust gas temperature sensor wiring harness P-clip from the rocker lever cover.



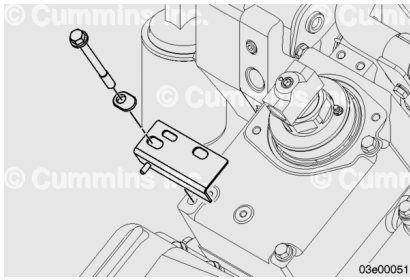
Remove the P-clip securing the exhaust gas temperature sensor wire to the injector cover plate.



Remove the two capscrews securing the injector cover plate to the rocker lever cover and remove the injector cover plate. Pry injector cover plate evenly on opposite sides using a heel bar. Remove the o-ring seal from the rocker lever cover plate and discard it.



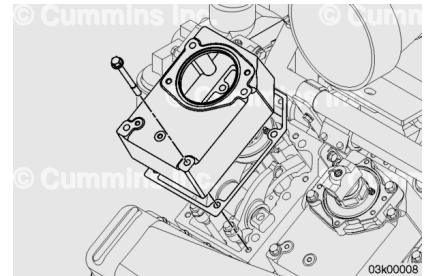
Remove the capscrew securing the wiring harness bracket to the rocker lever cover and remove the bracket.



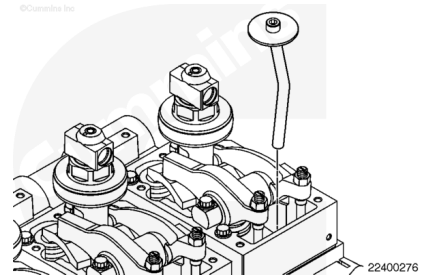
Remove the remaining three capscrews securing the rocker lever cover.

Remove the rocker lever cover and gasket.

Discard the old gasket.



Install the cylinder head protective cover, Part Number 4918282, in the hole located between the push tubes to prevent debris from entering the engine while the rocker lever cover is removed.

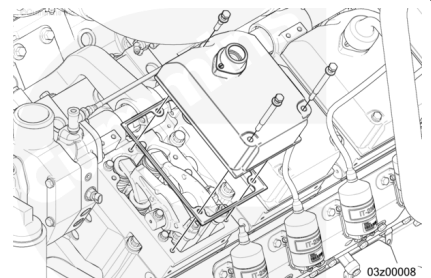


Natural Gas

Remove the capscrews from the rocker lever cover.

Remove the rocker lever cover and gasket.

Discard the gasket.



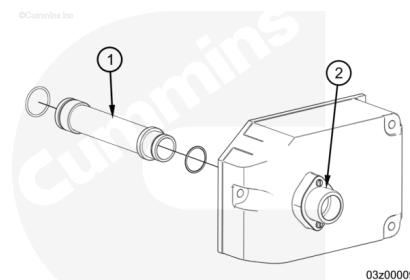
Disassemble

Natural Gas

Remove the tube adapter (1) from the rocker lever.

The tube cap (2) does **not** need to be removed unless it is damaged.

Remove and discard the o-rings.

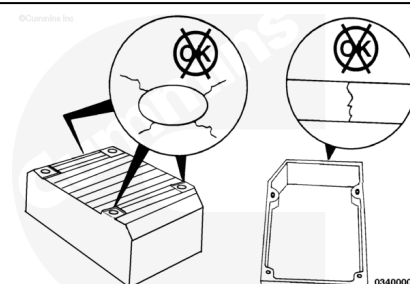


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.

Use solvent to clean the rocker lever covers.

Dry with compressed air.

Inspect the rocker lever cover for cracks.

with Electronically Actuated Injector

⚠ WARNING ⚠

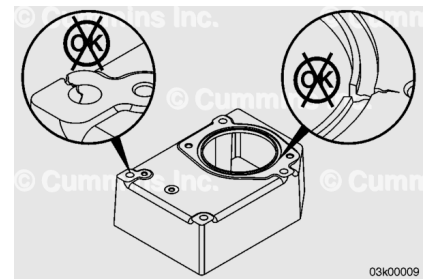
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

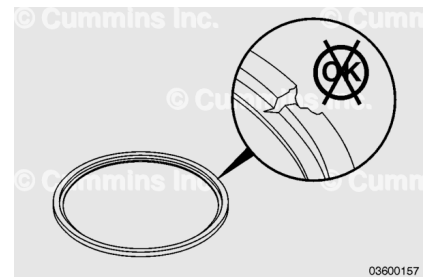
Use solvent to clean the rocker lever cover and dry with compressed air.

Inspect all parts for cracks, damaged or stripped threads, and damaged capscrew holes.



Inspect the injector cover plate for cracks, damage, stripped threads, or damaged holes. Replace if damaged.

Inspect the injector cover plates dust seal for signs of tears or damage. Replace if damaged.



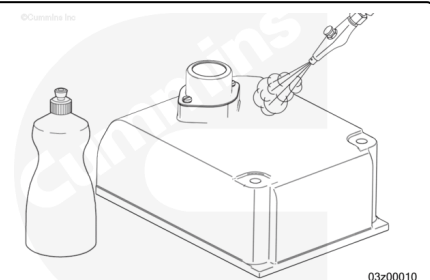
Natural Gas

⚠ WARNING ⚠

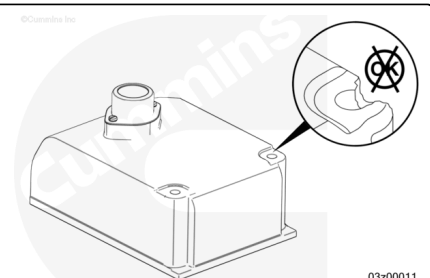
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the rocker lever cover with a strong solution of detergent.

Dry with compressed air.



Inspect the rocker lever cover for cracks or other damage.

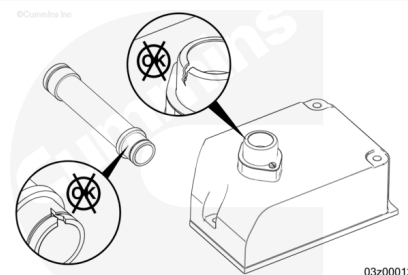


Inspect the adapter tube for cracks, bends, or other damage.

If damaged, the adapter tube **must** be replaced.

Inspect the sealing tube cap for cracks or other damage.

If damaged, the tube cap **must** be replaced.



Assemble

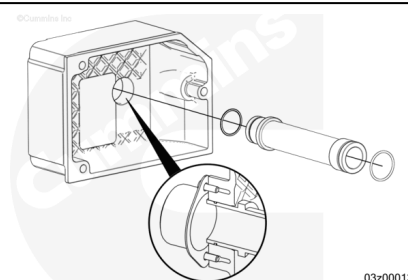
Natural Gas

Install the tube cap, if removed.

Install the tube adapter on the rocker lever cover.

Install new o-rings.

The boss on the tube **must** fit tightly against the underside of the rocker lever cover.



Install

with Mechanically Actuated Injector

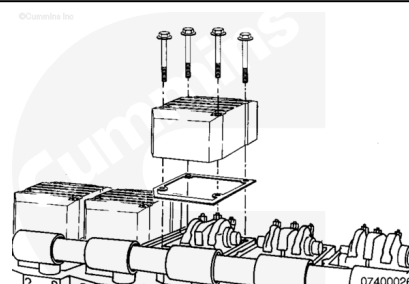
⚠ CAUTION ⚠

Do not use gasket cement. Gasket cement will prevent the gasket from sealing properly.

Install the gaskets, rocker lever covers, and capscrews.

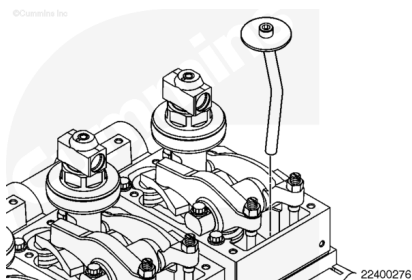
Tighten all the capscrews.

Torque Value: 45 n•m [33 ft-lb]



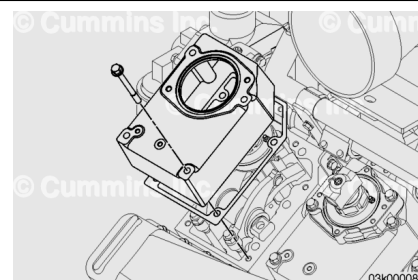
with Electronically Actuated Injector

Remove the cylinder head protective cover from the push tube hole.



⚠ CAUTION ⚠

Do not use gasket cement. Gasket cement will prevent the gasket from sealing properly.



Install the new rocker lever cover gasket and the rocker lever cover.

Install the first three rocker lever cover capscrews, leaving out the capscrew in the wiring harness bracket location.

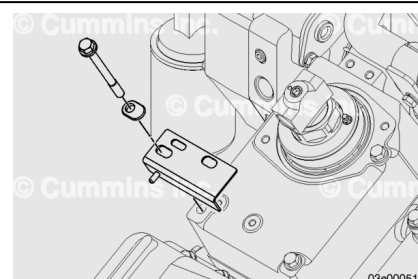
Hand-tighten the capscrews.

Install the wiring harness bracket, securing the capscrew and washer.

Make sure the holes in the bracket line up with the holes in the cover.

Tighten all four rocker housing cover capscrews.

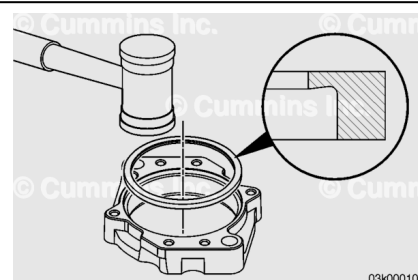
Torque Value: 40 n·m [30 ft-lb]



If the injector cover plates dust seal was damaged and needs to be replaced, lubricate the new dust seal with clean engine oil.

Check that the smooth side of the seal is facing downward and the side with the groove is facing upward.

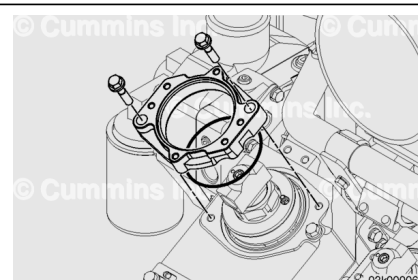
Use a soft rubber mallet to tap the seal into place.



Lubricate the large o-ring and damper on the injector with clean engine oil. The engine oil will protect the o-ring and damper during the installation of the rocker lever cover plate.

Install the new o-ring on the rocker lever cover plate.

Install the rocker lever cover plate on the injector with the seal facing outward, and tap it into place with a soft rubber mallet.

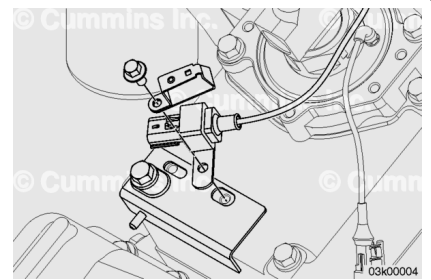


Install the rocker lever cover plate capscrews and tighten.

Torque Value: 40 n•m [30 ft-lb]

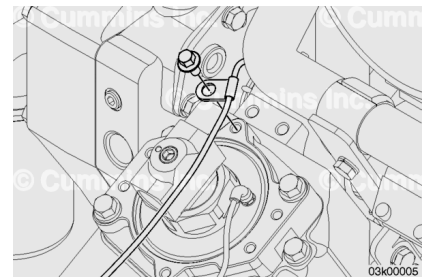
Install the injector wiring harness bracket and exhaust gas temperature sensor wiring harness P-clip to the rocker lever cover.

Torque Value: 20 n•m [177 in-lb]



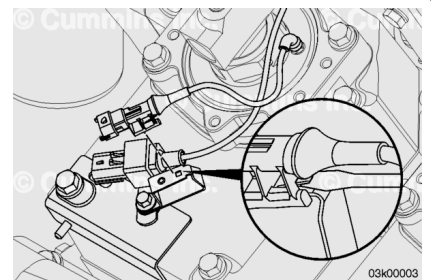
Install the P-clip securing the exhaust gas temperature sensor wire to the injector cover plate.

Torque Value: 20 n•m [177 in-lb]



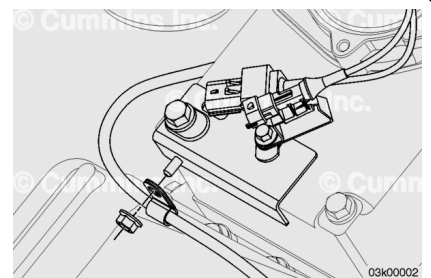
Install the injector wiring harness connector to the bracket by sliding it onto the bracket until the connector locks into position.

If the locking fixture on the connector is missing, use a cable tie, Part Number 3822924, or equivalent, to secure the wiring harness to the bracket.

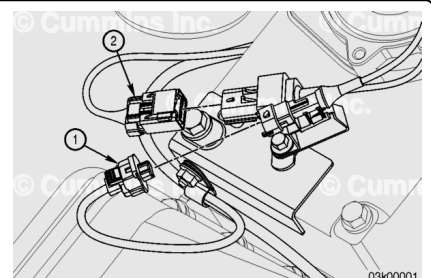


Install the P-clip securing the wiring harness to the rocker lever cover.

Torque Value: 20 n•m [177 in-lb]



Connect the electrical connectors for the injector (1) and exhaust gas temperature sensor (2).



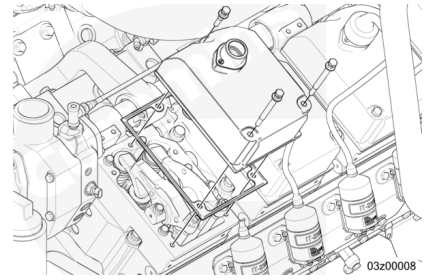
Natural Gas

Install the new gasket and rocker lever cover.

Install the capscrews from the rocker lever cover and hand tighten.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the air crossover. Refer to Procedure 010-020 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-020.html)
- Operate the engine and check for leaks.

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for leaks.

Natural Gas

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the upper portion of the air intake manifold lateral and vertical clamping brackets, if required.
- Install the spark plug wire (non-shielded). Refer to Procedure 013-014 in Section 13. (</qs3/pubsys2/xml/en/procedures/28/28-013-014.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.